# RESOURCE ALLOCATION IN DISTRIBUTED COMPUTING

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**2026**

# Statement of Originality

I hereby certify that the work embodied in this thesis is the result of original research, is free of plagiarised materials, and has not been submitted for a higher degree to any other University or Institution.

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Shangyi Shao

# Supervisor Declaration Statement

I have reviewed the content and presentation style of this thesis and declare it is free of plagiarism and of sufficient grammatical clarity to be examined. To the best of my knowledge, the research and writing are those of the candidate except as acknowledged in the Author Attribution Statement. I confirm that the investigations were conducted in accord with the ethics policies and integrity standards of Nanyang Technological University and that the research data are presented honestly and without prejudice.

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The contributions of the co-authors are as follows:

- A/Prof Schmid provided the initial project direction and edited the manuscript drafts.
- I prepared the manuscript drafts. The manuscript was revised by Dr Hester and Dr. Blanchard.
- I co-designed the study with A/Prof Siegbert Schmid and performed all the laboratory work at the School of Materials Science and Engineering and the Singapore Synchrotron Light Source. I also analyzed the data.
- All microscopy, including sample preparation, was conducted by me in the Facility for Analysis, Characterization, Testing and Simulation.
- Dr James Hester assisted in the collection of the neutron powder diffraction data.
- Dr Peter Blanchard assisted in the interpretation of the X-ray absorption spectroscopy data and carried out the spectral interpretation.
- Dr Wojciech Miiller assisted in the collection and provide guidance in the interpretation of the magnetic measurement data.

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The contributions of the co-authors are as follows:

- Prof Ting suggested the materials area and edited the manuscript drafts.
- I wrote the drafts of the manuscript. The manuscript was revised together with Dr. Sartbaeva and Dr. Yao.
- II performed all the materials synthesis, collected X-ray diffraction patterns and visible light spectra, carried transmission electron microscopy, and conducted data evaluation.
- IDr. Y. Fang conducted the Rietveld analysis of the powder X-ray diffraction data and single crystal structure determinations.
- IDr U. Hintermair conducted the molecular dynamics simulations.
- IMs. A. Sartbaeva prepared the samples for electron microscopy.

Mar. 2019


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Date

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Shangyi Shao

# Acknowledgements

I wish to express my greatest gratitude to my advisor, Assoc. Prof Xueyan Tang. Without his full support, it would be impossible for me to

*"If I had one hour to save the world, I would spend 55 minutes defining the problem and only five minutes finding the solution."*

—Einstein, Albert

To my dear family

# Abstract

The proliferation of computing-intensive workloads has firmly established geo-distributed architectures—spanning cloud platforms, interconnected data centers, and high-performance computing clusters—as the standard paradigm for modern large-scale data processing. To exploit data locality and minimize bandwidth consumption, these systems routinely distribute computing jobs into multiple localized tasks deployed across disparate geographical sites where the requisite input data reside. This framework has become highly prevalent across diverse application domains, including sensor networks, climate modeling, and large-scale machine learning. However, the heterogeneous nature of these environments introduces profound challenges in resource allocation and job scheduling. Due to the imbalanced distribution of workloads and varying site capacities, classical single-cluster scheduling mechanisms often precipitate resource bottlenecks, system inefficiencies, and severe job starvation. Furthermore, while geo-distributed sites are typically interconnected via well-provisioned networks, the strategic utilization of these network resources to transfer data and computational demands across sites remains a complex, NP-hard problem. More critically, a fundamental dichotomy exists between optimizing system efficiency (e.g., minimizing average job response time) and guaranteeing equitable resource allocation among competing jobs. Purely efficiency-driven algorithms often compromise fairness, while strict fairness-oriented policies inevitably degrade overall throughput. To address these multi-dimensional challenges, this dissertation systematically investigates resource allocation and job scheduling in geo-distributed systems, proposing a series of novel, network-aware scheduling frameworks that leverage inter-site bandwidth to synergistically optimize processing efficiency and rigorous fairness guarantees.

The first phase of this research focuses on redefining equitable resource allocation in distributed execution environments by integrating network transmission capabilities. Existing studies on fair resource allocation, such as Max-min Fairness and Dominant Resource Fairness, predominantly operate under single-site assumptions

where all systemic resources are homogeneously accessible. In geo-distributed systems, however, evaluating fairness solely at the localized level is inherently flawed; a resource-intensive job constrained to a low-capacity site may suffer from highly unfair allocations despite monopolizing local resources. To resolve this, this dissertation proposes Dynamic Aggregate Max-min Fairness (DAMF), a pioneering resource allocation policy that leverages spare inter-site network bandwidth to dynamically transfer data and computation demands. DAMF mandates that the aggregate resource allocation for each job across all computational sites achieves max-min fairness over the entirety of the system's demand transmission and allocation solution space. Theoretically, this research rigorously proves that the DAMF policy satisfies three critical axiomatic properties of fair division: Pareto efficiency, envy-freeness, and strategy-proofness. From a practical standpoint, the study develops comprehensive algorithms and a fast approximation tailored to compute optimal demand transmissions and precise DAMF resource allocations subject to dynamic demand distributions and strict network capacity constraints. Extensive experimental evaluations demonstrate that DAMF significantly outperforms conventional baseline policies, successfully achieving superior fairness metrics among competing jobs while simultaneously enhancing the aggregate utility of the distributed system.

Building upon the integration of network resources, the second phase of this dissertation pivots to the optimization of system efficiency, specifically targeting the minimization of average job response time in distributed environments. In geo-distributed architectures, a job's completion time is strictly dictated by the completion of its terminal task across all executing sites. Consequently, traditional efficiency-centric strategies like Shortest Remaining Processing Time (SRPT), which are optimal for single-site clusters, yield deeply suboptimal results in distributed settings. The problem is exacerbated by the intricate interdependencies between the local execution order of jobs and the cross-site transmission of tasks: transferring a single task can fundamentally alter the global optimal execution sequence, while a newly proposed execution order may necessitate a divergent transmission strategy. To disentangle this complexity, this research introduces Workload-Aware Selection with Transmission (WAST), an integrated job scheduling algorithm designed to concurrently compute the global execution order and optimal inter-site task transmissions. WAST systematically formulates the response-time minimization objective into a sequence of optimization problems, which are elegantly transformed into multiple minimum-cost flow problems solvable by advanced operational research solvers (e.g., OR-Tools, Cplex, GUROBI). To mitigate the substantial computational overhead inherent to the massive solution space, a highly efficient heuristic approximation algorithm is further proposed without significantly compromising scheduling fidelity. Driven by real-world production cluster traces from Alibaba, the empirical validation confirms that both WAST and its heuristic counterpart compute near-optimal execution sequences and transmission strategies, drastically reducing average job response times, particularly under highly skewed spatial workload distributions.

The final phase of the dissertation unifies the dual objectives by establishing a robust framework that successfully reconciles the inherent trade-off between scheduling efficiency and long-term fairness. Recognizing that instantaneous fairness can cripple system throughput and pure efficiency can induce prolonged job starvation, this research introduces a stringent long-term fairness criterion inspired by the concept of ”protective scheduling.” Under this criterion, a scheduling policy must guarantee that no job experiences a completion time later than it would under a baseline distributed Processor Sharing (PS) protocol, which theoretically guarantees equitable capacity sharing among all active jobs. To operationalize this, the dissertation formulates the distributed protective scheduling problem by constructing a mathematically rigorous slack system characterized by dual perspectives: a global view and an independent view. Crucially, mathematical proofs presented in this study reveal that, owing to the volatile nature of dynamic job arrivals and shifting load distributions, no purely globally order-based scheduling policy can maintain strict protectiveness. To transcend this theoretical limitation, a novel heuristic algorithmic framework, denoted as PFSP_x, is developed. PFSP_x acts as a hybrid scheduling engine that strictly enforces the protective fairness boundaries while aggressively optimizing the internal processing sequences to maximize efficiency. Trace-driven comparative analyses unequivocally demonstrate that PFSP_x effectively eliminates the violations observed in globally order-based extensions and highly aggressive efficiency algorithms (such as SRPT and SWAG). Ultimately, PFSP_x achieves the highest job processing efficiency among all rigorously protective policies, providing a low-cost, resilient scheduling paradigm for modern, large-scale geo-distributed computing systems.

# Contents

# List of Figures

# List of Tables

# Symbols and Acronyms

## Symbols

| | |
|---|---|
| $\mathcal{R}^n$ | the $n$-dimensional Euclidean space |
| $\mathcal{H}$ | the Euclidean space |
| $\|\cdot\|$ | the 2-norm of a vector or matrix in Euclidean space |
| $\|\cdot\|_G$ | the induced norm of a vector in G-space |
| $\|\cdot\|_E$ | the induced norm of a vector or matrix in probabilistic space |
| $\odot$ | the Hadamard (component-wise) product |
| $\otimes$ | the Kronecker product |
| $\langle\cdot,\cdot\rangle$ | the inner product of two vectors |
| $\circ$ | the composition of functions |
| | |
| $\nabla f$ | the gradient vector |
| $\mathcal{C}^k$ | the function with continuous partial derivatives up to $k$ orders |
| $x_{i,k}$ | the $i$-th component of a vector $x$ at time $k$ |
| $\bar{x}$ | the vector with the average of all components of $x$ as each element |
| $\mathbf{1}$ | all-ones column vector with proper dimension |
| $\mathcal{C}$ | the average space, i.e., $span\{\mathbf{1}\}$ |
| $\mathcal{C}^{\perp}$ | the disagreement space, i.e., $span^{\perp}\{\mathbf{1}\}$ |
| $\Pi_{\|}$ | the projection matrix to the average space $\mathcal{C}$ |
| $\Pi_{\perp}$ | the projection matrix to the disagreement space $\mathcal{C}^{\perp}$ |
| $O(\cdot)$ | order of magnitude or ergodic convergence rate (running average) |
| $o(\cdot)$ | non-ergodic convergence rate |
| $\mathcal{N}_i$ | the index set of the neighbors of agent $i$ |

# Acronyms

| | |
|---|---|
| DOP | Distributed Optimization Problem |
| EDOP | Equivalent Distributed Optimization Problem |
| SDOP | Stochastic Distributed Optimization Problem |
| OEP | Optimal Exchange Problem |
| OCP | Optimal Consensus Problem |
| DOCP | Dynamic Optimal Consensus Problem |
| | |
| AugDGM | Augmented Distributed Gradient Methods |
| AsynDGM | Asynchronous Distributed Gradient Methods |
| D-ESC | Distributed Extremum Seeking Control |
| D-SPA | Distributed Simultaneous Perturbation Approach |
| D-FBBS | Distributed Forward-Backward Bregman Splitting |
| ADMM | Alternating Direction Method of Multipliers |
| DSM | Distributed (Sub)gradient Method |
| | |
| GAS | Globally Asymptotically Stable |
| UGAS | Uniformly Globally Asymptotically Stable |
| SPAS | Semi-globally Practically Asymptotically Stable |
| USPAS | Uniformly Semi-globally Practically Asymptotically Stable |
| | |
| HoS | Heterogeneity of Stepsize |
| FPR | Fixed Point Residual |
| OBE | Objective Error |
| | |
| i.i.d. | independent and identically distributed |
| *a.s.* | almost sure convergence of a random sequence |

# Chapter 1

# Introduction

## 1.1 Research Scope

It is increasingly common for data-processing jobs in geo-distributed systems to spread tasks among multiple sites for data locality: each task will be deployed and processed at some site where their input data are available [? ? ? ]. Through this framework, the system can effectively save the bandwidth cost and improve the service response time. Geo-distributed processing has been prevalent for data analytics in various domains such as sensor networks [? ], climate science [? ? ] and stock exchange [? ]. In such systems, providing fair resource allocation to jobs can help with preventing starvation and fulfilling QoS for all jobs, but it is a non-trivial problem. The problem consists of two critical hurdles: how to decide the distributed resource allocation for each job with various amount of resource demand distribution across sites, and how to utilize the limited network resources to transfer tasks of certain jobs between sites to achieve more fair allocation and better overall utility.

Existing work on fair resource allocation has primarily focused on a cluster of machines at a single site, where all resources in the system are assumed to be available for processing all jobs. Typical examples include the classical Max-min Fairness [? ] and Proportional Fairness [? ]. As an extension, Constrained Max-min Fairness [? ] achieves max-min fair allocation given the placement constraints of jobs on the machines. Dominant Resource Fairness [? ] and Resource Elasticity Fairness [? ] present fair allocations of multiple resources. On the other hand, the study of distributed job execution has mainly focused on improving job scheduling efficiency or saving bandwidth cost [? ? ]. Little work has considered fair resource allocation among jobs executing over multiple sites. Recently, Guan et al. [? ] has explored the conventional max-min fairness in distributed job execution by requiring the aggregate resource allocation across all sites to be max-min fair. However, if the job workloads are distributed unevenly among the sites, the resulting resource allocation could be quite inefficient or unfair with respect to each job's demand. For example, if a high-demand job's entire workload is located at one site with small capacity, its allocation can still be unfair even if it takes up all resources at that site.

As different sites in the distributed system are often inter-connected with well-provisioned networks, the spare bandwidth can be utilized to transfer data and hence computation demands among sites to enhance the efficiency and improve the fairness of resource allocation among jobs. Since there is a huge solution space, the choice of moving which job's data from which site to another is a challenge. To the best of our knowledge, achieving fair resource allocation with exploiting the use of idle network resources has not been widely studied. In this paper, we apply max-min fairness to distributed job execution with utilization of network resources to transfer job demands between sites. We propose a resource allocation policy called Dynamic Aggregate Max-min Fairness (DAMF), which requires each job's aggregate resource allocation across all sites to be max-min fair among all possible demand transmissions and resource allocations. We prove that DAMF satisfies three important properties of fairness: Pareto efficiency, envy-freeness and strategy-proofness. Given the demand distributions of jobs and the network capacity constraints, we develop algorithms to compute the demand transmission and the DAMF resource allocation to each job at each site. Experimental results demonstrate that compared with existing allocation policies, DAMF can achieve resource allocations with better fairness among jobs and improve the system utility.

Meanwhile, this kind of geo-distributed computing systems introduces many new challenges, such as how to improve system efficiency and decrease the volume of data transmission. In a single data center scenario, the strategy for achieving the highest job processing efficiency (i.e., minimize the average job response time – the response time of a job is defined as the duration between its arrival time and its completion time) is to sort and execute jobs based on their computation loads (i.e., the total processing time of all tasks of each job) from low to high, known as the Shortest Remaining Processing Time (SRPT) strategy. Similar ideas can also be applied to the distributed job execution scenario where jobs are comprised of tasks to be processed at multiple data centers, but this can be far from optimal. In geo-distributed job execution, the completion time of each job is determined by the completion time of its last task across all sites. Therefore, the execution order of jobs within individual sites leaves a huge space for optimization, especially when the task distributions of jobs among different sites are imbalanced. How to achieve optimal job scheduling in distributed job execution is an NP-hard problem.

What makes the problem even more complicated is how to utilize network resources in the system to further enhance job processing efficiency. Since the completion time of the last task decides the response time of a job, transferring the input data of tasks to other sites can potentially bring forward the execution of tasks and thereby improve the job's response time. However, there exists complex interdependency between the execution order of jobs and the transmission of tasks. Even the transmission of a single task may change the optimal execution order of jobs, while different job execution orders may, in turn, require different task transmissions to minimize the average job response time. As a result, it is highly challenging to find the optimal solution.

In this paper, we focus on developing job scheduling algorithms which utilize network resources to transfer tasks and improve job processing efficiency in geo-distributed systems. We present a novel job scheduling strategy, called Workload-Aware Selection with Transmission (WAST), designed to optimize average job response time. WAST computes the global execution order of jobs and a set of task transmissions between different sites in an integrated manner. More specifically, WAST searches for the task transmission strategy to achieve the best possible response time of each job by formulating an optimization problem and transforming it into multiple minimum-cost flow problems that can be efficiently solved by popular solvers (e.g., OR-Tools [? ], Cplex [? ], and GUROBI [? ]), and iteratively schedules the job with the minimum response time until all jobs are scheduled. We also present a heuristic approximation algorithm that reduces the computational complexity without compromising the quality of scheduling much. We validate the effectiveness of the WAST algorithm and its approximation using publicly available job traces from Alibaba. Experimental results show that both algorithms can efficiently compute near-optimal job execution order and task transmissions, thereby significantly reducing job response times, especially in scenarios where the initial computation load distributions of jobs are highly skewed across sites.

## 1.2 Thesis Contributions

Chapter 1 introduces ...

Chapter 2 reviews ...

More chapters ....

....

# Part I

# Part Name: Use it when there are many chapters

# Chapter 2

# Literature Review

## 2.1 Part 1

When you cite a paper [1], the back reference from bibgraph will apper as page number.

You can also cite paper with author name using the command 'citet': such as: Bauschke and Combettes [1].

## 2.2 Part 2

cite another paper [2].

**Lemma 2.1** (My lemma)**.** *A great lemma.*

$$c^2 = a^2 + b^2 \tag{2.1}$$

**Theorem 2.2** (My theorem)**.** *A great theorem.*

$$c^2 = a^2 + b^2 \tag{2.2}$$

*Proof.* The proof is intuitive. □

**Corollary 2.3** (My corollary)**.** *A great corollary.*

$$c^2 = a^2 + b^2 \tag{2.3}$$

**Proposition 2.1** (My proposition)**.** *A great proposition.*

$$c^2 = a^2 + b^2 \tag{2.4}$$

**Example 2.1** (My example)**.** A great example.

$$c^2 = a^2 + b^2 \tag{2.5}$$

**Definition 2.1** (My definition)**.** A great definition.

$$c^2 = a^2 + b^2 \tag{2.6}$$

**Assumption 2.1** (My assumption)**.** A great assumption.

$$c^2 = a^2 + b^2 \tag{2.7}$$

*Remark* 2.1 (My remark)*.* A great remark.

$$c^2 = a^2 + b^2 \tag{2.8}$$

# Chapter 3

# Chapter3 Name

## 3.1 Section1

See Figure 3.1


FIGURE 3.1: Another illustration.

Let's cite out first table: Table 3.1.

| Table | Group 1 | | Group 2 | |
|---|---|---|---|---|
| | Col 1 | Col 2 | Col 1 | Col 2 |
| Row 1 | 14.37 | **5.76** | 2.65 | **2.84** |
| Row 2 | 5.43 | **7.36** | 2.22 | **2.49** |
| Row 3 | 5.54 | **5.68** | **4.42** | 2.92 |

TABLE 3.1: My Table.

# Part II

# Again for Second Part

# Chapter 4

# Chapter4 Name

Quisque facilisis auctor sapien. Pellentesque gravida hendrerit lectus. Mauris rutrum sodales sapien. Fusce hendrerit sem vel lorem. Integer pellentesque massa vel augue. Integer elit tortor, feugiat quis, sagittis et, ornare non, lacus. Vestibulum posuere pellentesque eros. Quisque venenatis ipsum dictum nulla. Aliquam quis quam non metus eleifend interdum. Nam eget sapien ac mauris malesuada adipiscing. Etiam eleifend neque sed quam. Nulla facilisi. Proin a ligula. Sed id dui eu nibh egestas tincidunt. Suspendisse arcu.

## 4.1 Section 1

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## 4.2 Section 2

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# Chapter 5

# Chapter5 Name

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## 5.1 Section 1

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## 5.2 Section 2

Quisque facilisis auctor sapien. Pellentesque gravida hendrerit lectus. Mauris rutrum sodales sapien. Fusce hendrerit sem vel lorem. Integer pellentesque massa vel augue. Integer elit tortor, feugiat quis, sagittis et, ornare non, lacus. Vestibulum posuere pellentesque eros. Quisque venenatis ipsum dictum nulla. Aliquam quis quam non metus eleifend interdum. Nam eget sapien ac mauris malesuada adipiscing. Etiam eleifend neque sed quam. Nulla facilisi. Proin a ligula. Sed id dui eu nibh egestas tincidunt. Suspendisse arcu.

# Appendix A

# Proofs for Part I or Chapter 3

## A.1 Proof of Lemma

$$\psi^{av}(\theta) = \frac{1}{T}\int_0^T [\psi(\theta + \mu(\tau)) + C] \otimes \frac{\mu(\tau)}{a} d\tau$$

## A.2 Proof of another Lemma

$$\begin{aligned} \gamma_1(\|x\|) \leq W(t, x) \leq \gamma_2(\|x\|) \\ \frac{\partial W}{\partial t} + \frac{\partial W}{\partial x}\phi(t, x, 0) \leq -\gamma_3(\|x\|) \end{aligned} \tag{A.1}$$

# List of Author's Awards, Patents, and Publications[1]

## Awards

- **Best Paper Awards**, "A Great System," *Nature*.

## Patents

- **A Great System**, "A Great System," *Nature*.

## Journal Articles

- **My name** and My colleague, "A Great System," *Nature*.

## Conference Proceedings

- **My name**, My colleague 1, My colleague 3 and My colleague 3, "Greater System," in *Conference of Vision, 2018*.

[1] The superscript * indicates joint first authors

# Bibliography

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